

DTSM: Digital Twin-Enabled Secure Multicast Signcryption Protocol for Low-Altitude UAV Networks

Luyao Liu¹, Jin Cao¹, Ruhui Ma¹, Boyu Deng¹, Guanjie Li¹, Yinghui Zhang¹, and Hui Li¹

Abstract—Digital Twins (DT) are increasingly becoming a core infrastructure for large-scale Unmanned Aerial Vehicle (UAV) swarm orchestration in low-altitude intelligent networks. By constructing a physical-virtual closed loop, UAVs frequently need to send multicast command messages based on DTs to numerous UAVs via insecure wireless links. Signcryption ensures secure data exchange by simultaneously guaranteeing confidentiality and authentication, thus better matching UAV performance. However, most state-of-the-art signcryption schemes optimize the receiver through aggregation or batch verification, while the cost of sending the same command to multiple receivers still increases rapidly with swarm size, creating a bottleneck of receiver-friendly but sender-costly. To overcome this limitation, we propose a digital twin-enabled secure multicast signcryption protocol (DTSM), which enables multicast messages to be delivered in signcryption form, allowing only designated UAVs to decrypt, authenticate, and execute them. We first construct a three-layer digital twin architecture, which further illustrates the role of DTs in collaborative tasks within UAV swarms. Then, we design a group key distribution mechanism based on polynomial interpolation and a secure multicast signcryption protocol, embedding the multicast key into a Lagrange polynomial, allowing each designated UAV to recover the key locally. Formal proofs under the random oracle model and protocol formal verification using Scyther demonstrate that the proposed DTSM achieves confidentiality, unforgeability, authentication, and resistance to replay attacks, while informal analysis indicates that it also supports conditional anonymity and traceability. Evaluation results show that the proposed DTSM significantly reduces the communication and computational overhead at the sending side compared with related schemes, making it suitable for large-scale receiving scenarios.

Index Terms—Digital twin, low-altitude network, multicast, signcryption, Unmanned Aerial Vehicle (UAV).

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I. INTRODUCTION

GLOBALLY, the low-altitude economy and Advanced Air Mobility (AAM) are transitioning from piloted to scalable operations. In Europe, the European Aviation Safety Agency (EASA)'s Easy Access Rules for U-space provide the binding digital traffic management baseline for routine drone operations [1]. In China, the four-ministry Implementation Plan for General Aviation Equipment Innovation and Application (2024–2030) positions the low-altitude economy as a strategic growth engine, with large-scale demonstrations and ecosystem development [2].

The Low-Altitude Intelligent Network integrates several core technological domains, including U-space/UTM for coordinated airspace and traffic management [3], autonomous Detect-And-Avoid (DAA) systems integrated with high-integrity navigation [4], and Integrated Sensing And Communication (ISAC). In particular, joint waveform and resource orchestration for multi-Unmanned Aerial Vehicle (UAV) operations is seen as a promising evolution toward 6G, enhancing both link reliability and environmental awareness [5].

On the application front, UAVs have emerged as core enablers of the low-altitude economy, undertaking a wide range of missions including logistics delivery, emergency response or medical supply delivery, urban governance, smart city operations, patrol inspection, surveying, environmental monitoring, as well as agricultural, forestry, pastoral, and fishery tasks [6]. UAVs are expected to experience continued expansion and ecosystem growth within the low-altitude economy [7].

To support complex multi-UAV operations, including multi-task coordination, formation cooperation, and dynamic adaptation, Digital Twin (DT) technology is increasingly integrated into low-altitude networks. DT enables virtual representation, closed-loop state awareness, and predictive optimization for task orchestration and resource allocation across UAV swarms [7]. For example, Tang et al. proposed a DT-enabled, mission-driven framework for formation scale management, member configuration, task crowdsourcing, and resource coordination [8]. Zhang et al. demonstrated in UAV and Mobile Edge Computing (MEC) scenarios that DT can reduce latency and energy consumption while improving task success rates, also proposing a collaborative management architecture for multi-formation systems [9]. In future Internet of Things (IoT) and networked systems, DT acts as an infrastructure-level enabler, supporting a priori

planning and closed-loop feedback for grouping and scheduling [10].

In DT-supported UAV scenarios, reliable and efficient communication is essential for exchanging control instructions, status updates, and sensory data. In real-world swarm missions, the command node may need to send the same control commands to a group of drones performing the same task, such as formation control, area inspection, or coordinated emergency response. In this case, the communication requirement is secure multicast transmission to multiple designated receivers sharing the mission context. Signcryption schemes are often adopted to achieve both confidentiality and authentication simultaneously. Recent advances in aggregate or multi-receiver signcryption have largely focused on the receiver side: through aggregation or batch verification, multiple messages can be verified at once, greatly reducing computational and communication overhead for the verifier [11], [12]. However, on the sender side, when the same command must be delivered to multiple UAVs, the cost of mainstream signcryption continues to increase rapidly with the number of receivers [13]. This results in a “receiver-friendly but sender-heavy” imbalance, which becomes a bottleneck for scalable swarm command and control. This imbalance is particularly pronounced when command nodes that require DT assistance repeatedly issue the same mission instructions to multiple designated drones in large-scale collaborative operations. This issue serves as the primary motivation for this paper to adopt a DT-driven grouping strategy to explicitly reduce sender-side overhead [11].

Existing works have demonstrated that DTs can be constructed in urban low-altitude airspace using real-time point clouds to achieve task assignment and trajectory optimization [14]. Furthermore, DTs directly support formation optimization, resource scheduling, and task offloading for drone swarms, thereby providing an engineering foundation for the “group-first-then-deliver” orchestration strategy proposed in this work [15]. Therefore, to alleviate the computational load on the sender, our scheme employs a signcryption protocol based on DT grouping for efficient multi-UAV command transmission. Specifically, our contributions are as follows.

- We propose a hierarchical three-layer digital twin architecture, including the following layers. (1) A Low-Altitude Cyber-Physical Execution Layer that serves as the physical foundation containing entities, responsible for real-time perception, data collection, and instruction execution. (2) A Digital Twin Network Layer that maps spatiotemporal data from physical entities into computable DTs to achieve state alignment, predictive simulation, and conflict detection within the virtual space. (3) A Digital Twin Coordination Layer functioning as the central brain for closed-loop optimization, which integrates the twin mirror, predictive analysis, strategy generation, and security interaction modules to orchestrate strategies and enforce security governance.
- We propose a digital twin-enabled secure multicast Signcryption protocol for low-altitude UAV networks. A polynomial interpolation-based key distribution mechanism is designed, utilizing Lagrange interpolation to construct group keys and ensuring that only designated receivers can recover the key and decrypt it.

- Our protocol achieves optimization of the sender’s overhead. When the sender multicasts the same instruction to multiple receivers, the proposed scheme substantially reduces the sender-side communication and computational overhead, significantly reducing the load. The communication and computational costs of the proposed scheme are compared with those of mainstream certificateless signcryption schemes. The results show that the proposed scheme effectively reduces sender-side overhead in the considered multi-receiver scenario.
- Under the Random Oracle Model (ROM), formal proofs demonstrate that the scheme satisfies confidentiality (IND-CCA), unforgeability (EUF-CMA), and resistance to replay attacks, with security advantages bounded based on the ECDLP/ECDHP hardness assumptions. The Scyther tool is used to formally verify the security abilities, including mutual authentication, message synchronization, and key confidentiality. The informal analysis indicates that the scheme possesses conditional anonymity and traceability.

In summary, by integrating DT technology, we design an efficient multicast signcryption protocol suitable for low-altitude networks. It significantly improves efficiency while ensuring security, providing a feasible scheme for future UAV swarm cooperative communication in 6G environments.

The remainder of this paper is organized as follows. Section II reviews related work. Section III introduces preliminary knowledge. Section IV elaborates the system model, threat model, and security model. Section V details the proposed protocol. Security analysis and performance evaluation are presented in Sections VI and VII, respectively. Finally, Section VIII concludes the paper.

II. RELATED WORK

In DT-supported UAV scenarios, reliable and efficient communication is essential for exchanging control instructions, status updates, and sensory data. Signcryption schemes are commonly employed to achieve simultaneous confidentiality and authentication. Therefore, in this section, we present an overview of existing signcryption schemes. These schemes have found widespread application in mainstream scenarios such as Vehicular Ad-hoc Networks (VANETs) and have been successfully adapted for resource-constrained IoT environments. Moreover, batch verification and aggregate verification schemes have been developed to reduce the computational overhead on the receiver side significantly. Furthermore, there are also authentication and signcryption schemes for multicast scenarios to achieve the authentication and secure data transmission.

A. Certificateless Signcryption Schemes in VANETs

Christian et al. [16] proposed a lightweight certificateless signcryption scheme for secure VANET data exchange using hyperelliptic curve cryptography and edge computing. However, the scheme’s practicality is limited by challenges in cross-platform compatibility and large-scale deployment. Liu et al. [17] introduced a pairing-free certificateless signcryption scheme operating in the standard model, providing provably

secure confidentiality and unforgeability for secure data communication in VANETs. However, the signature verification process in this scheme incurs high computational complexity due to its reliance on multiple hash operations. Zhan et al. [18] introduced an efficient multi-mode certificateless ring signcryption scheme for VANETs to enable dynamic privacy-level adjustment. This scheme supports three privacy modes—full anonymity, linkable anonymity, and conditional traceability—under a single algorithmic framework. However, the mode-switching mechanism faces potential recognition conflicts in dynamic network topologies.

B. Certificateless Signcryption Schemes in the IoT

Dadkhah et al. [19] introduced a provably secure certificateless signcryption scheme within the ROM for secure IoT data communication, achieving low computational overhead for both confidentiality and unforgeability. However, the security of this scheme relies on the idealized random oracle assumption and is not supported by a proof in the standard model. Furthermore, a potential risk of partial key escrow exists in its key generation process. Kasyoka et al. [20] introduced a certificateless signcryption scheme utilizing Gaussian likelihood censored regression and a Rabin-based Jaccard index for secure, energy-efficient data transmission in Wireless Sensor Networks (WSNs), reporting a 93% packet delivery ratio and a 15% reduction in energy consumption. However, the scheme depends on centralized key generation and the computationally intensive Jaccard index calculation for signature verification, which could lead to significant latency accumulation under high load conditions. Jiang et al. [21] introduced a certificateless signcryption-based supervised anonymous transaction model for blockchain account models, designed to balance user privacy (strong anonymity and confidentiality) with regulatory compliance. However, the threshold shared custody of the regulatory key introduces a risk of recovery latency in scenarios involving dynamic changes among the supervising members. Ali et al. [22] introduced a lightweight certificateless signcryption with equality test scheme using hyperelliptic curves for Wireless Body Area Networks (WBANs), which supports equality testing on encrypted medical data in the cloud. However, this functionality depends on the active upload of user trapdoors. Additionally, trapdoor transmission in multi-hop network environments could pose a significant communication bottleneck. Bai et al. [23] introduced a module lattice-based certificateless online/offline signcryption scheme for Internet of Medical Things (IoMT) environments, designed to provide quantum resistance and IND-CCA/EUF-CMA security, with an online phase requiring only 2 polynomial-vector additions. However, the offline phase necessitates pre-computation of high-dimensional polynomial matrix multiplications. Moreover, the practical optimization of the module lattice parameters presents non-trivial challenges.

C. Signcryption Schemes Featuring Batch/Aggregate Verification

Xie et al. [24] introduced a lightweight anonymous certificateless multi-receiver signcryption scheme to enable secure

broadcast communication for resource-constrained IoT devices. The scheme provides mutual anonymity between sender and receivers, supports multiple receivers, and employs pairing-free lightweight computations. However, its ciphertext length expands linearly as the number of receivers increases dynamically. Chiang et al. [25] proposed an efficient certificateless signcryption scheme in VANETs. This scheme supports both anonymous mutual authentication between vehicles (V2V) and efficient batch authentication between vehicles and infrastructure (V2I). By integrating certificateless cryptography with signcryption technology, it achieves these security goals while avoiding computationally intensive elliptic curve pairing operations, thereby significantly reducing both computational overhead and communication latency. Yang et al. [12] introduced a pairing-free certificateless online/offline signcryption batch verification scheme for Video-to-Radar (V2R) communication in VANETs. The scheme is provably secure in the random oracle model but faces two main challenges. On the one hand, its node selection mechanism based on Gaussian likelihood censored regression could cause energy-aware delays in high-dynamic environments. On the other hand, its batch verification aggregation is susceptible to a full-batch rejection attack if malicious inputs are included. Wang et al. [26] introduced a pairing-free, blockchain-assisted certificateless aggregate signcryption scheme designed for secure V2I communication in VANETs. The scheme provides provable IND-CCA2 and EUF-CMA security in the ROM. However, its security is contingent upon the hardness of the ECDLP and ECDH problems. Additionally, storing public keys on the blockchain may lead to potential latency during key retrieval.

D. Authentication and Signcryption Schemes for Multicast or Multi-Receiver Scenarios

Peng et al. [27] proposed a certificateless multi-message multi-receiver signcryption (CLMMSC) scheme for multicast communication in edge computing environments, which achieves efficient data confidentiality, unforgeability, and receiver anonymity. However, the scheme involves multiple scalar multiplications and modular operations during the signcryption phase, resulting in a higher computational cost compared to other comparable schemes. Wang et al. [28] proposed a broadcast authentication method based on certificateless public key cryptography (CL-PKC) for VANETs, aiming to achieve identity authentication, confidentiality, integrity, and non-repudiation of data transmission between vehicles and multiple roadside units (RSUs). However, this protocol relies on traditional authenticated encryption with key agreement, which increases signaling overhead and latency in data transmission scenarios. Zhong et al. [29] proposed a cross-domain batch signcryption authentication method for secure multi-domain vehicle communication in VANETs, leveraging blockchain and dynamic accumulators. However, the communication reliability of this approach depends on stable connections between roadside units (RSUs) and multi-access edge computing (MEC) servers, which may compromise real-time authentication efficiency. Gong et al. [30] proposed the Efficient Certificateless Group Signcryption-Based Homomorphic (ECGSH) scheme for secure data transmission in

IIoT scenarios. The scheme implements an efficient group signcryption mechanism that combines certificateless signcryption with homomorphic encryption, enabling data processing in an encrypted state. However, its performance is suboptimal for data transmissions below 50 bytes, which may limit its applicability in ultra-low data rate applications. Additionally, its dynamic key update mechanism may experience synchronization latency during rapid topology changes. Su et al. [31] proposed a personalized multi-receiver certificateless signcryption scheme with cross-domain authentication for IoV, achieving customized data sharing for multiple specific receivers together with conditional anonymity and traceability. However, the blockchain-assisted cross-domain authentication framework may incur additional system complexity.

While these schemes achieve varying degrees of success in balancing security and efficiency, a fundamental limitation persists in current batch verification paradigms: the computational and communication costs on the sender side increase rapidly with the number of receivers, and this aspect has not yet been adequately addressed in existing literature.

III. PRELIMINARIES

This section fixes the mathematical background and notation used throughout the paper, including the elliptic-curve setting, the computational assumptions, authenticated encryption with associated data, polynomials and interpolation over finite fields, and the hash-function interfaces.

A. Elliptic-Curve Setting

Let $E/\mathbb{F}_p$ be an elliptic curve over a prime field $\mathbb{F}_p$ with short Weierstrass form.

$$E(\mathbb{F}_p) : y^2 \equiv x^3 + ax + b \pmod{p}, \quad (1)$$

where $a, b \in \mathbb{F}_p$, $4a^3 + 27b^2 \not\equiv 0 \pmod{p}$. We fix a prime-order cyclic subgroup $\mathbb{G} \subseteq E(\mathbb{F}_p)$ of order q with generator P . Group operations are written additively; scalar multiplication is denoted xP for $x \in \mathbb{Z}_q$.

The security arguments rely on standard hardness assumptions in $\mathbb{G}$:

- 1) Elliptic-Curve Discrete Logarithm Problem (ECDLP): Given (P, X) with $X = xP$ for an unknown $x \in \mathbb{Z}_q$, it is computationally infeasible to recover x .
- 2) Elliptic-Curve Computational Diffie–Hellman Problem (ECDHP): Given (P, xP, yP) for random $x, y \in \mathbb{Z}_q$, it is computationally infeasible to compute xyP by any probabilistic polynomial-time algorithm.

B. Authenticated Encryption With Associated Data

Authenticated Encryption with Associated Data (AEAD) is a symmetric primitive that provides both confidentiality and ciphertext integrity. An AEAD scheme consists of two polynomial-time algorithms AEAD.Enc , AEAD.Dec . Given a key K , a nonce $N \in \{0, 1\}^{\lambda_N}$, a plaintext M , and associated data AD , it outputs

$$(C, \text{Tag}) = \text{AEAD.Enc}_K(N, M; AD), \quad (2)$$

and decryption returns

$$\text{AEAD.Dec}_K(N, C, \text{Tag}; AD) \in \{M, \perp\}. \quad (3)$$

The associated data AD is authenticated but not encrypted, and is used to bind context to the ciphertext.

The security of AEAD schemes relies on the following two standard properties:

- 1) Indistinguishability under Chosen-Plaintext Attack (IND-CPA): the ciphertext reveals no information about the plaintext.
- 2) Ciphertext Integrity (INT-CTXT): it is computationally infeasible for an adversary to generate a valid ciphertext–tag pair without knowing the encryption key.

Indistinguishability under Chosen Ciphertext Attack (IND-CCA): The combination of these two properties implies chosen-ciphertext security, $\text{IND-CPA} \wedge \text{INT-CTXT} \Rightarrow \text{IND-CCA}$ [32].

C. Polynomials Over Finite Fields

Let q be the prime order of the cyclic group $\mathbb{G}$ introduced above. A (univariate) polynomial of degree d over $\mathbb{Z}_q$ is

$$f(x) = c_0 + c_1x + c_2x^2 + \cdots + c_dx^d, \quad (4)$$

where $c_i \in \mathbb{Z}_q$ for $i = 0, \dots, d$ and $c_d \neq 0$. In cryptographic applications, the constant term c_0 often embeds a secret or session material, while higher-order coefficients act as randomizers.

a) *Key properties:*

- 1) Degree: The largest exponent with a nonzero coefficient equals $\deg(f) = d$.
- 2) Evaluation: For any $x \in \mathbb{Z}_q$, the value $f(x)$ is efficiently computable.
- 3) Interpolation: Given $d+1$ pairs $(x_j, f(x_j))$ with distinct $x_j \in \mathbb{Z}_q$, the polynomial f is uniquely determined.

b) *Lagrange interpolation:*

Given t distinct points $(x_1, y_1), \dots, (x_t, y_t)$ in $\mathbb{Z}_q \times \mathbb{Z}_q$, the unique polynomial f of degree at most $t-1$ that satisfies $f(x_j) = y_j$ is

$$f(x) = \sum_{j=1}^t y_j \cdot \prod_{\substack{1 \leq m \leq t \\ m \neq j}} \frac{x - x_m}{x_j - x_m} \in \mathbb{Z}_q[x]. \quad (5)$$

Remark: Whenever group elements need to serve as x -coordinates in later interpolation, we first map them to field elements in $\mathbb{Z}_q$ via a hash-to-field function and then interpolate over $\mathbb{Z}_q$.

D. Hash Functions and Notation

We model the following seven independent hash functions as random oracles with implicit encoding $\langle \cdot \rangle$ and domain separation: $H_0 : \{0, 1\}^* \rightarrow \{0, 1\}^\ell$, $H_1 : \{0, 1\}^* \times \mathbb{G}^2 \rightarrow \mathbb{Z}_q^*$, $H_2 : \{0, 1\}^* \rightarrow \mathbb{Z}_q$, $H_3 : \mathbb{G} \rightarrow \mathbb{Z}_q$, $H_4 : \mathbb{G} \rightarrow \mathbb{Z}_q$, $H_5 : \{0, 1\}^* \rightarrow \mathbb{Z}_q$, $H_6 : \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$. Here $\mathbb{Z}_q^*$ denotes nonzero residues modulo q . ℓ denotes the length of an identity.

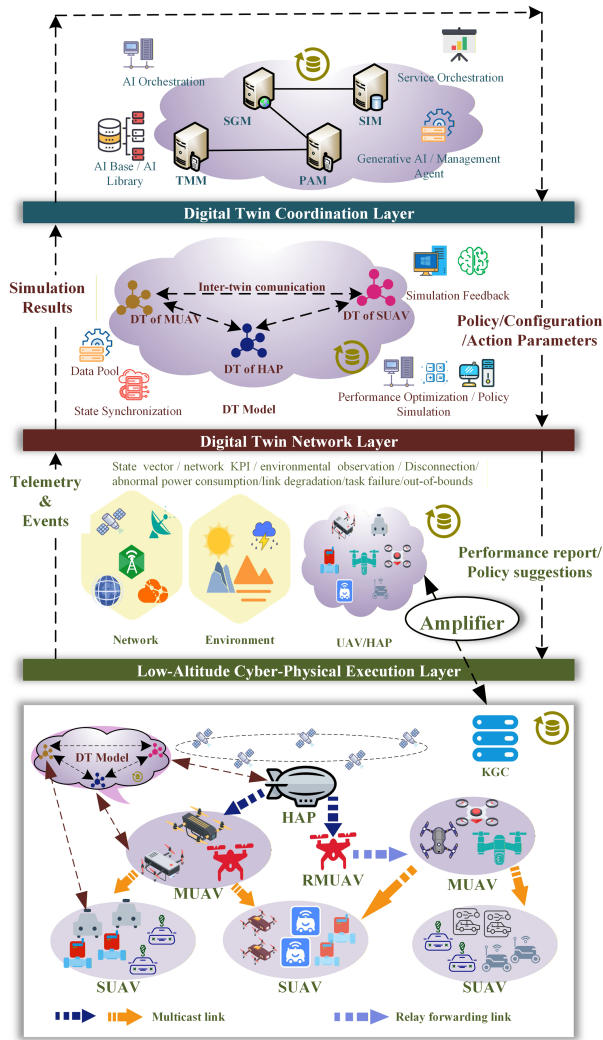


Fig. 1. Hierarchical Three-layer Digital Twin Architecture for Low-Altitude UAV Networks.

IV. SYSTEM AND SECURITY MODEL

A. System Model

We first design a Hierarchical Three-Layer Digital Twin Architecture for Low-Altitude UAV Networks. Each layer involves multiple entities, each with distinct responsibilities to ensure secure communication, mission decision-making, and service execution. As shown in Fig. 1, the system components at each layer are defined as follows.

1) *Low-Altitude Cyber-Physical Execution Layer*: This layer corresponds to the physical entities in the low-altitude network, serving as both the physical carrier and execution endpoint. It is responsible for real-world perception and actuation, continuously collecting status and event data from UAVs, network components, and the environment. The acquired physical states are transmitted back to their corresponding DTs via secure channels. The proposed architecture is shown in the enlarged panel in Fig. 1 and comprises the following four entities: KGC (Key Generation Center), HAP (High Altitude Platform), MUAV / RMUAV (Medium Unmanned Aerial Vehicle / Relay MUAV), and UAV (Small Unmanned Aerial Vehicle).

- 1) *KGC*: Responsible for generating and distributing keys to all entities in the physical execution layer. It publishes common system parameters and handles physical entity registration and key issuance. In practical applications, the 5G Core Network Element Unified Data Management (UDM) can act as the KGC [33]. Since the UDM resides in the operator-controlled 5G core security domain, it provides a natural trusted execution environment for key-management functionalities. In particular, the 5G security specification requires that long-term authentication keys and privacy-related home-network private keys be protected within the UDM security environment and not leave it in an unprotected form. Therefore, we assume the KGC instantiated in the UDM is well protected by the 5G core security infrastructure, and the registration interaction with the KGC is conducted over a protected channel [34].
- 2) *HAP*: A persistent hovering platform (such as a stratospheric airship, solar-powered drone, or high-altitude balloon) deployed in the 20-100 km near-altitude airspace serves as the communication and mission hub for the entire system [35], connecting the Digital Twin Network Layer with the Low-Altitude Cyber-Physical Execution Layer, enabling efficient operation of the DT-driven low-altitude drone system architecture. The HAP's long-term airborne capability (months to years), payload capacity of hundreds of kilograms, and low-cost deployment offer unique advantages in future integrated air-space-ground-sea and low-altitude networks, including wide-area coverage, long-duration high-altitude operations, and highly flexible deployment [36]. Its deep collaboration with satellites, drones, and ground base stations is driving the evolution of near-altitude communications from relay nodes to intelligent near-altitude computing networks [35], [36], [37].
- 3) *MUAV/RMUAV*: Regional Manager or Mission Relay Node. It acts as a group manager, receiving multicast commands from the HAP and assigning tasks to the SUAVs under its control. MUAV can also act as a relay RMUAV, helping HAP forward messages when multicast commands cannot be delivered to MUAV.
- 4) *SUAV*: Mission-performing UAVs. They receive instructions from MUAVs or directly from the HAP and perform specific tasks, such as inspection, transportation, and emergency rescue.

2) *Digital Twin Network Layer*: This layer maps spatiotemporal data from the physical world into computable DTs, achieving state alignment, scenario replay, and predictive simulation. It evaluates the effectiveness of different strategies and generates performance reports/strategy recommendations for decision-making. Each physical entity has a corresponding DT in this layer [38]. Physical entities upload their status and sensory data to the DTs via secure channels [39]. These DTs provide state synchronization, task simulation, and information collaboration in the virtual space. The DT entities are interconnected through secure channels, exchanging information such as state updates, capability declarations, conflict detection, and draft candidate

receiver sets. The interconnection results are submitted to the collaboration layer for final decision-making.

3) *Digital Twin Coordination Layer*: Based on simulation evaluation results, this layer performs strategy generation, orchestration, and closed-loop optimization, uniformly distributes strategy/configuration/action parameters, and adaptively iterates based on feedback. It acts as the “central brain” of the system. While it does not directly control physical devices, it provides unified data integration, prediction and evaluation, security governance, and strategy generation for each DT. Subsequently, it distributes executable strategies and security materials to the corresponding DTs, which then drive execution at the physical layer. The coordination layer consists of the following four modules.

- 1) *TMM (Twin Mirror Module)*: It aggregates and aligns the states and events from various DTs to form a consistent “trusted view.” It performs temporal alignment, handles anomalies and missing data, and triggers resampling or rollback recommendations when necessary.
- 2) *PAM (Predictive Analysis Module)*: It leverages this trusted view to conduct task and network side predictive evaluations (such as link quality, load, latency, energy consumption, and timeout risk), and derives constraints and preferences for strategy formulation.
- 3) *SGM (Strategy Generation Module)*: Based on these constraints, it performs system-level scheduling and strategy generation, including receiver set selection, resource orchestration, and task assignment. The output is securely encapsulated by the security module before being distributed.
- 4) *SIM (Security Interaction Module)*: It provides authentication, authorization, integrity assurance, and key management for collaborative layer interactions, ensuring that strategies are securely and accurately delivered to the corresponding DTs while supporting auditability and traceability.

The Low-Altitude Cyber-Physical Execution Layer is responsible for reporting its operational status and executing multicast instructions. The Digital Twin Network Layer, serving as a one-to-one virtual mapping of each physical entity, receives real-time feedback from the physical layer and enables interoperability among DTs. Building on this foundation, the Digital Twin Coordination Layer performs predictive analysis and comprehensive evaluation, generates unified security policies, and distributes them to the corresponding DTs, thereby forming a closed-loop system that integrates the virtual and physical realms. This architecture ensures that physical entities are accurately mirrored in the virtual space, where real-time data enables continuous online analysis and optimized decision-making. The results are then fed back to the physical side, consistently enhancing mission completion rates, operational reliability, and overall system efficiency.

B. Adversary Model

In this paper, we use the Dolev–Yao (DY) model as a threat model. First proposed by Dolev and Yao in 1983, the DY model symbolically (ideal cryptography) characterizes the strongest

active network adversary [40]. It assumes that the attacker has complete control over the public channel and can schedule and drive multiple rounds of protocol interactions concurrently. Therefore, it is particularly suitable for evaluating the robustness of protocols at the message flow and key binding levels.

Under the DY model, the attacker is granted complete control over the public channel and can intercept, modify, delete, delay, and replay any message. They can also forge legitimate identities and inject self-made messages into the network, thereby actively interfering with protocol execution [40]. In the symbolic description, the attacker’s knowledge of message items is closure-accessible, allowing for synthesis and decomposition, enabling attacks such as man-in-the-middle, replay, and concurrent session interleaving [41]. At the same time, the model treats cryptographic primitives as “perfect black boxes.” Without the key, it is impossible to recover plaintext from ciphertext, nor is it possible to forge signatures or MACs [42]. Hashes and random numbers are often abstracted as ideal functions and sources of freshness [43]. Therefore, the analysis within the DY model focuses on logical layers such as protocol messaging and identity/session binding, rather than underlying computational cracking capabilities.

C. Security Model

This section details the security model used in the analysis. All proofs are performed within the Random Oracle Model (ROM), where hash functions are modeled as independent random oracles. We describe the underlying computational setup and the attacker’s capabilities, then define their oracle access and freshness conditions, and finally formalize confidentiality, integrity, reliability, and replay protection objectives, along with their corresponding dominance symbols. Furthermore, we also informally discuss two security properties: conditional anonymity and traceability.

Computational setting: Let $(\mathbb{G}, +, P)$ be a cyclic subgroup of an elliptic curve of prime order q , with scalar field $\mathbb{Z}_q$. Group operations are written additively, and aP denotes scalar multiplication for $a \in \mathbb{Z}_q$. The security relies on the hardness of the ECDLP and ECDHP in $\mathbb{G}$. Hash functions $H_0, \dots, H_6$ are modeled as independent ROMs. The symmetric encryption is implemented using the Authenticated Encryption with Associated Data(AEAD) scheme, and AEAD.Enc is assumed to be IND-CPA (Indistinguishability under Chosen-Plaintext Attack) and INT-CTXT (Integrity of Ciphertexts). Following the standard inference Ciphertext integrity and CPA security together imply CCA security, that is $\text{IND-CPA} \wedge \text{INT-CTXT} \Rightarrow \text{IND-CCA}$ [32].

Adversary: The adversary $\mathcal{A}$ is a probabilistic polynomial-time (PPT) algorithm that fully controls the public channel: it can eavesdrop, inject, delete, reorder, and replay messages, and may interleave multiple sessions concurrently. It may statically or adaptively corrupt non-challenge entities to obtain their long-term secrets. Physical side-channel leakage and denial-of-service aspects are not considered in this model.

Interfaces: Depending on the security experiment, $\mathcal{A}$ can access: (i) adaptive queries to the random oracles $H_0, \dots, H_6$;

(ii) procedure calls that generate honest transcripts; (iii) decryption/verification or signing/tagging oracles; and (iv) key-reveal interfaces on non-challenge sessions to capture forward and backward secrecy.

Freshness: A test (challenge) instance is said to be *fresh* if, before the testing phase, no conflicting corruption, key-reveal, or state-leakage operation has occurred on the tested party or session. Non-challenge instances may be revealed as needed to express AKE-style key indistinguishability.

Security goals: We formalize four orthogonal goals through indistinguishability and unforgeability games:

i) *Confidentiality* — indistinguishability of payloads against chosen-ciphertext attacks (IND-CCA) and (group) session keys (ROM + ECDHP + IND-CPA + INT-CTXT);

ii) *Integrity and source authentication* — ciphertext integrity (INT-CTXT) from AEAD and tag/signature unforgeability (EUF-CMA) under ROM + ECDLP;

iii) *Authorization/Soundness* — only designated receivers can reconstruct the correct key or plaintext from public transcripts;

iv) *Replay Resistance* — session freshness and per-session binding prevent cross-round, cross-session, and cross-domain replay acceptance.

Advantage notation: The corresponding advantages of $\mathcal{A}$ in each game are denoted by Adv^{conf} , Adv^{int} , $\text{Adv}^{\text{forge}}$, $\text{Adv}^{\text{sound}}$, and $\text{Adv}^{\text{replay}}$. Upper bounds are expressed in terms of oracle query counts and the underlying hard problems, for example:

$$\text{Adv}^{\text{conf}} \leq \text{Adv}_{\text{AEAD}}^{\text{IND-CPA}} + \text{Adv}_{\text{AEAD}}^{\text{INT-CTXT}} + \text{Adv}^{\text{ECDHP}} + \sum_k \frac{q_{H_k}}{2^{\lambda_k}},$$

where q_{H_k} is the number of queries to oracle H_k and λ_k is the effective entropy of its output domain, if directly mapped to $\mathbb{Z}_q$, the term can be written as q_{H_k}/q .

Additional Security properties: Besides the above four formal goals, we further discuss two properties at an informal level: (i) *Conditional anonymity*: Without the KGC master secret or authorized tracing capability, an outside adversary cannot efficiently distinguish which real identity corresponds to a valid pseudonym; (ii) *Traceability*: The authorized tracing entity can recover the real identity corresponding to a valid pseudonym, thereby supporting accountability when necessary.

V. OUR PROPOSED SCHEME: DTSM

This section introduces the two phases of our proposed protocol: entity registration phase and digital twin-enabled multicast signcryption phase. The main notations used in the proposed scheme and their corresponding descriptions are shown in Table I.

A. Entity Registration Phase

1) System Initialization:

Let $\mathbb{G}$ be an additive cyclic group of prime order q generated by P . The Key Generation Center (KGC) selects a master secret $s \in \mathbb{Z}_q^*$ and computes the master public key $P_{\text{pub}} = sP$. The system's public parameters are $(\mathbb{G}, q, P, P_{\text{pub}}, H_0, H_1, H_2, H_3, H_4, H_5, H_6)$, where the master secret s is securely kept by the KGC.

TABLE I
DEFINITION OF NOTATION

Notation	Definition
s	Master secret key selected by the KGC
P_{pub}	Master public key, computed as sP
U_n	An entity (e.g., HAP, MUAV, SUAV)
ID_n	Identity of entity U_n
PID_n	Anonymous identity for U_n
PK_n	Public key of U_n , the tuple (X_n, Y_n)
SK_n	Secret key of U_n , the tuple (x_n, y_n)
M	Mission message
GK	Group key
δ_i	Authentication tag computed by HAP
δ_{ii}	Relay-verifiable tag
M_{agg}	Aggregated mission message from MUAV to SUAVs
GK_{new}	New group key generated by MUAV
δ_j	Authentication tag computed by MUAV

2) Entity Registration:

The entity obtains the public and private keys through the following process.

- 1) Each entity (HAP/MUAV/RMUAV/SUAV) U_n chooses a private scalar $x_n \in \mathbb{Z}_q^*$ and computes its public component $X_n = x_n P$. U_n sends the registration request $\langle X_n, ID_n \rangle$ to the KGC through a secure channel.
- 2) The KGC samples $r_n \in \mathbb{Z}_q^*$ for each U_n and computes $Y_n = r_n P$. It also assigns a task-context identifier Ctx_n to the current registration instance, and generates the anonymous identity $PID_n = ID_n \oplus H_0(Ctx_n \parallel P_{\text{pub}} \parallel sX_n)$ for the U_n . Compute the binding challenge $h_n^* = H_1(PID_n, Y_n, P_{\text{pub}})$. Form the Schnorr-like response $y_n = r_n + s \cdot h_n^* \pmod{q}$. Then, the KGC masks the response as $y'_n = y_n + H_2(PID_n \parallel (sX_n)) \pmod{q}$. Finally, it returns $\langle Y_n, h_n^*, y'_n, PID_n, Ctx_n \rangle$ to the U_n . Given (PID_n, Ctx_n, X_n) , the KGC can compute $ID_n = PID_n \oplus H_0(Ctx_n \parallel P_{\text{pub}} \parallel sX_n)$ to trace the real identity.
- 3) The U_n verifies the pseudonym by computing $PID'_n \stackrel{?}{=} ID_n \oplus H_0(Ctx_n \parallel P_{\text{pub}} \parallel x_n P_{\text{pub}})$. Then it computes $\tau_n = H_2(PID_n \parallel (x_n P_{\text{pub}}))$, and checks the group equation $y'_n P \stackrel{?}{=} Y_n + h_n^* P_{\text{pub}} + \tau_n P$. If the equation holds, the U_n removes the blinding and derives its final partial key $y_n = y'_n - \tau_n \pmod{q}$. The resulting public/secret keys of U_n are $PK_n = (X_n, Y_n)$, $SK_n = (x_n, y_n)$. And it calculates $z_n = x_n + y_n$. Otherwise, the response is rejected.

B. Digital Twin-Enabled Multicast Signcryption Phase

During this phase, as shown in Fig. 2, the HAP, acting as the global command node, first leverages the real-time virtual mapping provided by DT technology to comprehensively analyze the location, payload, and link quality of each MUAV in the network. Based on the DT's predictions of coverage gaps and path risks, it dynamically selects highly reliable nodes to form an optimal

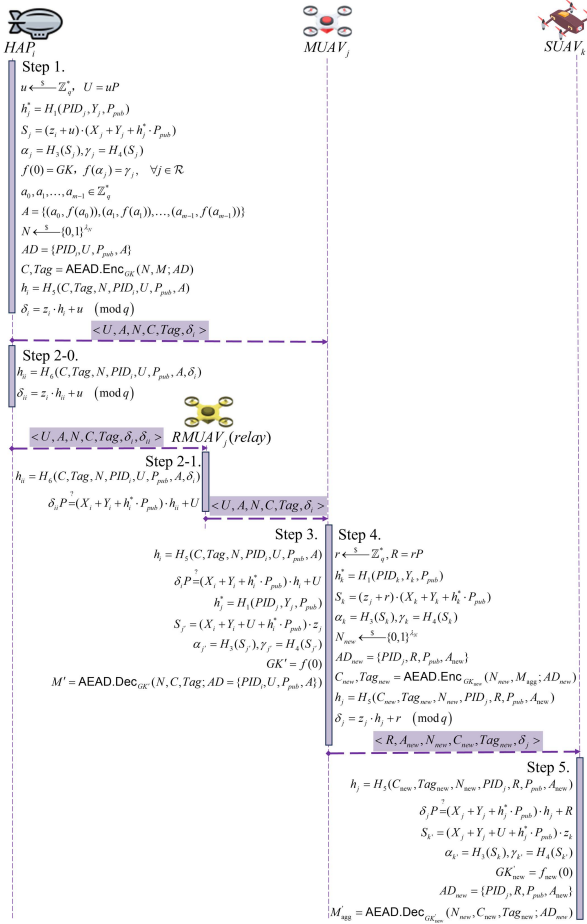


Fig. 2. Digital Twin-Enabled Multicast Signcryption Phase.

receiver set [44], [45], [46]. Next, the HAP employs a key distribution mechanism based on polynomial interpolation and symmetric encryption to ensure the confidentiality of multicast messages. Specifically, the HAP constructs a polynomial whose value at zero is the group key. It then encrypts the mission message using the key GK to produce a ciphertext, ensuring that only the target MUAV can decrypt it. If the MUAV that the HAP needs to select is too far away or exceeds the current multicast range due to other factors, it will select a closer RMUAV to act as a relay for forwarding.

After receiving the multicast, the MUAV first verifies the legitimacy of the message. Once verified, it reconstructs the polynomial using Lagrange interpolation to recover the group key. The ciphertext then decrypts the original mission message. Having successfully received the command, the MUAV prepares for subsequent processing or forwarding. If the RMUAV receives the message, it verifies the message's source and validity and forwards the message to the corresponding MUAV accordingly.

The specific process is described as follows.

Step 1: The HAP prepares a mission message M and a group key GK . It autonomously computes a policy, typically formulated as a decision set comprising target receivers, task parameters, and security rules. This policy is dynamically generated by the DT to ensure precise delivery and secure control

of multicast messages. Based on the current DT snapshot, the HAP first determines whether the selected MUAVs are within its effective multicast range and meet the link-quality threshold. If direct delivery is feasible, the HAP executes the following operations (Items 1–8), multicasts directly to the MUAVs, and then the protocol jumps to **Step 3**. If at least one target MUAV is out of range or the link quality is insufficient, the HAP executes Items 1–7 below, then switches to the relay path by performing **Step 2-0**, which is followed by **Step 2-1**. The HAP works as follows.

- 1) Sample $u \in \mathbb{Z}_q^*$ and sets $U = uP$.
- 2) Compute $h_j^* = H_1(PID_j, Y_j, P_{pub})$, anchor point $S_j = (z_i + u) \cdot (X_j + Y_j + h_j^* \cdot P_{pub}) \in \mathbb{G}$ for each designated receiver index $MUAV_j$, then derive $\alpha_j = H_3(S_j)$ and $\gamma_j = H_4(S_j) \in \mathbb{Z}_q$.
- 3) Suppose the HAP intends to deliver to a receiver set $\mathcal{R} = \{j_1, \dots, j_m\}$.
- 4) Compute the anchor pair (α_j, γ_j) for each $j \in \mathcal{R}$. Together with the distinguished point $(0, GK)$, this yields $m + 1$ points. There exists a unique polynomial $f : \mathbb{Z}_q \rightarrow \mathbb{Z}_q$ of degree at most m such that $f(0) = GK$ and $f(\alpha_j) = \gamma_j$ for all $j \in \mathcal{R}$.
- 5) Sample $a_0, \dots, a_{m-1} \in \mathbb{Z}_q^*$ uniformly at random and form the helper set $A = \{(a_i, f(a_i))\}_{i=0}^{m-1}$. To ensure that interpolation is well defined, the sender enforces that all a_i s are pairwise distinct; if any duplicate is found during generation, the repeated value is re-sampled.
- 6) Choose a fresh AEAD nonce $N \xleftarrow{\$} \{0, 1\}^{\lambda_N}$. Let $AD = (PID_i, U, P_{pub}, A)$. Compute the AEAD ciphertext-tag pair $(C, Tag) = \text{AEAD.Enc}_{GK}(N, M; AD)$.
- 7) Compute $h_i = H_5(C, Tag, N, PID_i, U, P_{pub}, A)$ as plaintext-level tag and set $\delta_i = z_i \cdot h_i + u \pmod{q}$.
- 8) Multicast $\langle U, A, N, C, Tag, \delta_i \rangle$ to the $MUAV_j$ and jump to **Step 3**.

Step 2-0: According to the policy decision in **Step 1**, some target MUAVs are located at a considerable distance or fall outside the current multicast range (or suffer from poor link quality), the HAP resorts to relay-assisted delivery. In this case, after completing Items 1–7 of **Step 1**, it selects a closer RMUAV as a relay and performs the following operations. The HAP works as follows.

- 1) Calculate $h_{ii} = H_6(C, Tag, N, PID_i, U, P_{pub}, A, \delta_i)$, the outer tag $\delta_{ii} = z_i \cdot h_{ii} + u \pmod{q}$, where δ_{ii} is verifiable by any relay $RMUAV_j$ using C, Tag and the inner tag δ_i is verified at the target domain.
- 2) Multicast $\langle U, A, N, C, Tag, \delta_i, \delta_{ii} \rangle$ to the relay $RMUAV_j$.

Step 2-1: The relay RMUAV receives the mission message from the HAP and verifies its legitimacy. It then forwards the message to the designated MUAV. The RMUAV works as follows:

- 1) Upon receiving $\langle U, A, N, C, Tag, \delta_i, \delta_{ii} \rangle$, perform $h_{ii} = H_6(C, Tag, N, PID_i, U, P_{pub}, A, \delta_i)$, and verify the $\delta_{ii} P \stackrel{?}{=} (X_i + Y_i + h_i^* \cdot P_{pub}) \cdot h_{ii} + U$.
- 2) If the check is valid, forward unchanged $\langle U, A, N, C, Tag, \delta_i \rangle$ to the MUAV in the target domain as indicated

by policy and jump to **Step 3**. No reconstruction of GK is attempted. Otherwise, the RMUAV discards the message and reports the verification failure together with local link status to its corresponding DT node. The DT node further exchanges this abnormal event with other twins via the coordination layer, so that subsequent relay selection and multicast policies can be adjusted accordingly, and this protocol instance is terminated locally.

Step 3: Upon receiving a multicast message, the MUAV verifies its validity and reconstructs the polynomial using Lagrange interpolation to recover the group key. Then, it decrypts the original task message and verifies it. The MUAV executes the specific process as follows. For a current designated MUAV receiver $j \in \mathcal{R}$, the sender identity is i .

- 1) Compute $h_i = H_5(C, Tag, N, PID_i, U, P_{pub}, A)$ and verify the tag $\delta_i P \stackrel{?}{=} (X_i + Y_i + h_i^* \cdot P_{pub}) \cdot h_i + U$.
- 2) Compute the anchor point $(\alpha_{j'}, \gamma_{j'})$, where $S_{j'} = (X_i + Y_i + U + h_i^* \cdot P_{pub}) \cdot z_j \in \mathbb{G}$, $\alpha_{j'} = H_3(S_{j'})$ and $\gamma_{j'} = H_4(S_{j'}) \in \mathbb{Z}_q$.
- 3) Before interpolation, the receiver checks that $\alpha_{j'} \notin \{a_0, \dots, a_{m-1}\}$; if a collision occurs, the current instance is rejected. Recover the interpolation polynomial $f(x)$ using $\{(\alpha_{j'}, \gamma_{j'})\}$ and A , recover $GK' = f(0)$, and decrypt $M' = AEAD.Dec_{GK'}(N, C, Tag; AD)$, where $AD = (PID_i, U, P_{pub}, A)$ is reconstructed from the received transcript.

Step 4: As a regional coordinator, the MUAV must process multiple batches of commands, potentially from multiple HAPs or even from the same HAP. After successfully decrypting multicast messages from the HAP, the MUAV re-aggregates and encapsulates the multiple task messages that it receives. Using DT intelligence, the MUAV intelligently selects target SUAV groups to redistribute and efficiently distribute tasks. From global commands to local execution, tasks are precisely aligned with regional resources. The MUAV works as follows.

- 1) Choose $r \in \mathbb{Z}_q^*$, set $R = rP$.
- 2) Suppose the MUAV intends to deliver to a receiver set $\mathcal{G} = \{k_1, \dots, k_m\}$. For each local SUAV index k , compute $h_k^* = H_1(PID_k, Y_k, P_{pub})$, $S_k = (z_j + r) \cdot (X_k + Y_k + h_k^* \cdot P_{pub}) \in \mathbb{G}$, then derive $\alpha_k = H_3(S_k)$ and $\gamma_k = H_4(S_k) \in \mathbb{Z}_q$.
- 3) Define f_{new} from $(0, GK_{new})$ and m $\{(\alpha_k, \gamma_k)\}$. Sample $a_0, \dots, a_{m-1} \in \mathbb{Z}_q^*$ uniformly at random and form the helper set A_{new} . The sender similarly enforces that all helper abscissas in A_{new} are pairwise distinct, and re-samples any repeated value if necessary.
- 4) Choose a fresh AEAD nonce $N_{new} \xleftarrow{\$} \{0, 1\}^{\lambda_N}$. Let $AD_{new} = (PID_j, R, P_{pub}, A_{new})$. Compute the AEAD ciphertext-tag pair $(C_{new}, Tag_{new}) = AEAD.Enc_{GK_{new}}(N_{new}, M_{agg}; AD_{new})$.
- 5) Compute $h_j = H_5(C_{new}, Tag_{new}, N_{new}, PID_j, R, P_{pub}, A_{new})$, $\delta_j = z_j \cdot h_j + r \pmod{q}$.
- 6) Multicast once $\langle R, A_{new}, N_{new}, C_{new}, Tag_{new}, \delta_j \rangle$ to all SUAVs in group.

Step 5: Each SUAV receives the message from the MUAV and performs the following process. For a current designated SUAV receiver $k \in \mathcal{G}$, the sender identity is j .

- 1) Compute $h_j = H_5(C_{new}, Tag_{new}, N_{new}, PID_j, R, P_{pub}, A_{new})$ and verify $\delta_j P \stackrel{?}{=} (X_j + Y_j + h_j^* \cdot P_{pub}) \cdot h_j + R$.
- 2) Compute $S_{k'} = (X_j + Y_j + R + h_j^* \cdot P_{pub}) \cdot z_k \in \mathbb{G}$, then derive $\alpha_{k'} = H_3(S_{k'})$ and $\gamma_{k'} = H_4(S_{k'}) \in \mathbb{Z}_q$, and check that $\alpha_{k'} \notin \{a_0, \dots, a_{m-1}\}$; if a collision occurs, the current instance is rejected. Reconstruct f_{new} from $\{(\alpha_{k'}, \gamma_{k'})\} \cup A_{new}$, and restore $GK'_{new} = f_{new}(0)$.
- 3) Let $AD_{new} = (PID_j, R, P_{pub}, A_{new})$, reconstructed from the received transcript, decrypt $M'_{agg} = AEAD.Dec_{GK'_{new}}(N_{new}, C_{new}, Tag_{new}; AD_{new})$.
- 4) If it is valid, execute the mission(s) specified in M_{agg} and report the execution result to the corresponding DT node. The DT node further exchanges this status with other twins through the coordination layer, where the aggregated feedback is used to update the global state and influence subsequent receiver grouping, path planning, and multicast policy generation. Otherwise, discard M'_{agg} , report the verification failure and local status to the DT node, and terminate this session locally without updating any mission state; the reported anomaly likewise participates in the above feedback process.

For each newly generated multicast instance, the sender must sample a new temporary scalar value $u/r \xleftarrow{\$} \mathbb{Z}_q^*$ and a new AEAD nonce $N/N_{new} \xleftarrow{\$} \{0, 1\}^{\lambda_N}$. The value u/r is for one-time use and must not be reused in different multicast sessions. Retransmission operations do not trigger a new sign-encryption or encryption process when they only correspond to repeated delivery of the same protected packet during transmission; however, any sender-initiated re-sending after time-out or lack of response is treated as a new multicast instance and must use fresh u/r values and AEAD nonces. In implementation, the sender should employ a standards compliant Random Bit Generator (RBG), a DRBG instantiated from an approved entropy source, with sufficient entropy input, protected internal state, and backtracking resistance [47], [48], [49].

In this scheme, the HAP, acting as the global command node, first leverages DT technology to dynamically analyze network status (such as node load and link quality) to generate an optimal set of receivers. It then encapsulates task messages using a polynomial interpolation-based encryption mechanism, attaches verification tags, and multicasts them to the target domain MUAVs. Upon receiving the messages, the MUAVs verify message integrity, decrypt the messages to obtain the original instructions, and then aggregate and redistribute multiple tasks based on local DT data, generating new group keys and policies, which are then securely forwarded to a selected group of SUAVs. Finally, the SUAVs decrypt the messages and execute the tasks, with the results fed back to the DT network layer, forming a closed-loop "physical-virtual" control loop. Each physical entity uploads status, results, and other information to the DT network in real time. The entire process relies on DT technology to dynamically map the virtual and physical worlds. Through collaborative decision-making based on cryptography and DTs, the efficient and secure flow of tasks from global planning to local execution is achieved.

C. Dynamic Group Update and Group Key Update

In DT-based low-altitude missions, the receiver group depends on the mission and may change during execution due to mobility, link fluctuations, or policy updates. Therefore, it is necessary to dynamically update the receiver group members and the Group Key in mid-mission.

Let the current receiver group be $\mathcal{R}$ and the updated group be $\mathcal{R}^u$. If a new receiver j^+ joins the group, then the updated set becomes $\mathcal{R}^u = \mathcal{R} \cup \{j^+\}$. If an existing receiver j^- leaves the group or is revoked, then the updated set becomes $\mathcal{R}^u = \mathcal{R} \setminus \{j^-\}$. We denote the size of the updated receiver set by $m^u = |\mathcal{R}^u|$.

The HAP performs the following steps to generate a fresh key for $\mathcal{R}^u$:

- 1) Sample a fresh group key GK^u and a fresh random scalar $u^u \xleftarrow{\$} \mathbb{Z}_q^*$, and set $U^u = u^u P$.
- 2) For each receiver $j \in \mathcal{R}^u$, compute $h_j^* = H_1(PID_j, Y_j, P_{pub})$ and the corresponding anchor $S_j^u = (z_i + u^u) \cdot (X_j + Y_j + h_j^* P_{pub}) \in \mathbb{G}$, then derive $\alpha_j^u = H_3(S_j^u) \in \mathbb{Z}_q$ and $\gamma_j^u = H_4(S_j^u) \in \mathbb{Z}_q$.
- 3) Define the unique polynomial $f^u(\cdot)$ of degree at most m^u such that $f^u(0) = GK^u$, $f^u(\alpha_j^u) = \gamma_j^u \forall j \in \mathcal{R}^u$.
- 4) Sample helper abscissas $a_0, \dots, a_{m^u-1} \in \mathbb{Z}_q^*$ and form $A^u = \{(a_\ell, f^u(a_\ell))\}_{\ell=0}^{m^u-1}$. To avoid degeneracy, the HAP enforces that all a_ℓ are pairwise distinct and $a_\ell \neq 0$. If any duplicate a_ℓ is detected, only the repeated positions are resampled until the constraint holds.
- 5) Compute $h^u = H_5(GK^u, PID_i, U^u, P_{pub}, A^u)$, $\delta^u = z_i \cdot h^u + u^u \pmod{q}$, and multicast the refresh message $\langle U^u, A^u, \delta^u \rangle$ to all receivers in $\mathcal{R}^u$.

Upon receiving $\langle U^u, A^u, \delta^u \rangle$, each MUAV $j \in \mathcal{R}^u$ proceeds as follows:

- 1) Compute $h_i^* = H_1(PID_i, Y_i, P_{pub})$ and derive $S_j^u = (X_i + Y_i + U^u + h_i^* P_{pub}) \cdot z_j \in \mathbb{G}$, then compute $\alpha_j^u = H_3(S_j^u)$ and $\gamma_j^u = H_4(S_j^u)$.
- 2) Reconstruct $f^u(\cdot)$ from $\{(\alpha_j^u, \gamma_j^u)\} \cup A^u$ and recover $GK^{u'} = f^u(0)$.
- 3) Compute $h^{u'} = H_5(GK^{u'}, PID_i, U^u, P_{pub}, A^u)$ and check $\delta^u P \stackrel{?}{=} (X_i + Y_i + h_i^* P_{pub}) \cdot h^{u'} + U^u$. If the equation holds true, accept and set GK^u as the new group key; otherwise, reject this refresh instance.

For a revoked/leaving node $j^- \notin \mathcal{R}^u$, the HAP does not generate any $(\alpha_{j^-}^u, \gamma_{j^-}^u)$ for the new instance. Hence, given only the public helper set A^u , the excluded node cannot reconstruct $f^u(\cdot)$ or recover GK^u .

Each update to the sender requires only one multicast message of (U^u, A^u, δ^u) , and each legitimate receiver only needs to perform interpolation recovery once.

VI. SECURITY ANALYSIS

In this section, we evaluate the security of the proposed scheme through formal analysis. We first demonstrate its correctness via mathematical derivation, followed by security proofs using the Random Oracle Model (ROM) and verification with the Scyther tool.

A. Correctness

In our scheme, the accuracy of the receiver's authentication of the sender depends on the consistency of the computation results between the two parties. Our proposed scheme ensures that the computation results are consistent for both the receiver and the sender, thus ensuring the security of communication between the two parties.

$$\begin{aligned}
 S_j &= (X_i + Y_i + U + h_i^* \cdot P_{pub}) \cdot z_j \\
 &= (z_i - s \cdot h_i^* + u + s \cdot h_i^*) \cdot P \cdot z_j \\
 &= (z_i + u) \cdot z_j \cdot P \\
 &= (z_i + u) \cdot (X_j + Y_j + h_j^* \cdot P_{pub}) \\
 S_k &= (X_j + Y_j + R + h_j^* \cdot P_{pub}) \cdot z_k \\
 &= (z_j - s \cdot h_j^* + r + s \cdot h_j^*) \cdot P \cdot z_k \\
 &= (z_j + r) \cdot z_k \cdot P \\
 &= (z_j + r) \cdot (X_k + Y_k + h_k^* \cdot P_{pub}) \\
 \delta_i P &= (z_i \cdot h_i + u) \cdot P \\
 &= z_i P \cdot h_i + uP \\
 &= (X_i + Y_i + h_i^* P_{pub}) \cdot h_i + U \\
 \delta_j P &= (z_j \cdot h_j + r) \cdot P \\
 &= z_j P \cdot h_j + rP \\
 &= (X_j + Y_j + h_j^* \cdot P_{pub}) \cdot h_j + R
 \end{aligned}$$

B. Formal Proofs in the Random Oracle Model (ROM)

In this subsection, we present a provable security analysis of the DTSM under the Random Oracle Model (ROM). We focus on two core steps, *Step 1* and *Step 3*. This analysis is based on the general security model described in Section IV-C.

1) *Notation and Assumptions*: All cryptographic assumptions, group and hash follow the conventions of Section IV-C and are briefly restated here for clarity of reference.

Hash functions $H_0, \dots, H_6$ are independent random oracles. The symmetric layer is instantiated with an AEAD scheme (Section III-B) and is assumed to satisfy IND-CPA confidentiality and INT-CTXT integrity under nonce uniqueness. By the standard implication $\text{IND-CPA} \wedge \text{INT-CTXT} \Rightarrow \text{IND-CCA}$ [32], payload confidentiality further holds against chosen-ciphertext attacks. Moreover, both ECDLP and ECDHP are assumed hard in $\mathbb{G}$.

The sender's public-key binding relation is

$$X_i + Y_i = z_i P - h_i^* \cdot P_{pub}. \quad (6)$$

2) *Protocol elements used in Step 1 and Step 3*: As described in Section V-B:

- The sender produces a one-shot Schnorr-like tag

$$\begin{aligned}
 h_i &= H_5(C, Tag, N, PID_i, U, P_{pub}, A) \in \mathbb{Z}_q, \\
 \delta_i &= z_i \cdot h_i + u \pmod{q}, \quad (7)
 \end{aligned}$$

and broadcasts $(U, A, N, C, Tag, \delta_i)$.

- A receiver j first recomputes $h_i = H_5(C, Tag, N, PID_i, U, P_{pub}, A)$ and verifies δ_i via

$$\delta_i P \stackrel{?}{=} (X_i + Y_i + h_i^* \cdot P_{pub}) \cdot h_i + U. \quad (8)$$

then if the check passes, it locally recomputes S_j , derives $\alpha_j = H_3(S_j)$ and $\gamma_j = H_4(S_j) \in \mathbb{Z}_q$, reconstructs f from $\{(\alpha_j, \gamma_j)\} \cup A$ (Lagrange interpolation) and retrieves $GK = f(0)$, then computes $M' = \text{AEAD.Dec}_{GK}(N, C, Tag; AD)$ with $AD = (PID_i, U, P_{pub}, A)$, it accepts only if $M' \neq \perp$.

3) Confidentiality of GK and M :

[IND-CPA for the HAP multicast processing]

An *external* adversary $\mathcal{A}$ (that never obtain any designated pair (α_j, γ_j)) chooses two equal-length messages (M_0, M_1) . The challenger runs Step 1 to produce (U, A, N^*, δ_i) and sets $AD^* = (PID_i, U, P_{pub}, A)$. It then computes $(C^*, Tag^*) = \text{AEAD.Enc}_{GK^*}(N^*, M_b; AD^*)$ for a random bit $b \in \{0, 1\}$, where $GK^* = f^*(0)$ for a fresh degree- $\leq m$ polynomial f^* constructed as in Step 1. The challenge transcript given to $\mathcal{A}$ is $(U, A, N^*, C^*, Tag^*, \delta_i)$. Finally, $\mathcal{A}$ outputs b' . Its advantage is

$$\text{Adv}_{\text{DTSM}}^{\text{IND-CPA}}(\mathcal{A}) = \left| \Pr[b' = b] - \frac{1}{2} \right|.$$

Theorem 6.1 (Confidentiality of GK and M): In the ROM, under ECDHP in $\mathbb{G}$ and the IND-CPA security of AEAD,

$$\text{Adv}_{\text{DTSM}}^{\text{IND-CPA}}(\mathcal{A}) \leq \text{Adv}_{\text{AEAD}}^{\text{IND-CPA}} + \text{Adv}^{\text{ECDHP}} + \frac{q_{H_3} + q_{H_4}}{q},$$

where q_{H_3} and q_{H_4} are the numbers of H_3 and H_4 queries by $\mathcal{A}$, respectively.

Furthermore, since the instantiated AEAD scheme satisfies both IND-CPA and INT-CTXT, the proposed construction also achieves outsider IND-CCA confidentiality, namely

$$\begin{aligned} \text{Adv}_{\text{DTSM}}^{\text{IND-CCA}}(\mathcal{A}) &\leq \text{Adv}_{\text{AEAD}}^{\text{IND-CPA}} + \text{Adv}_{\text{AEAD}}^{\text{INT-CTXT}} \\ &\quad + \text{Adv}^{\text{ECDHP}} + \frac{q_{H_3} + q_{H_4}}{q}. \end{aligned}$$

Proof sketch:

i) *Information-theoretic hiding of GK from A :* Since f has degree at most m , $m+1$ points are necessary and sufficient to determine f , and hence to determine $f(0) = GK$. The broadcast set A contains only m points and omits both $(0, GK)$ and every (α_j, γ_j) . Hence $f(0)$ is a priori hidden given only A .

ii) *Computational inaccessibility of (α_j, γ_j) :* To compute any $\alpha_j = H_3(S_j)$ and $\gamma_j = H_4(S_j)$, the adversary must derive S_j from a Diffie-Hellman style product that mixes (a) the HAP's fresh u or the sender's secret z_i with (b) the receiver's secret z_j . More precisely, $S_j = (z_i + u) \cdot (X_j + Y_j + h_j^* \cdot P_{pub})$. Since $X_j + Y_j + h_j^* \cdot P_{pub} = z_j P$, we have $S_j = (z_i + u) z_j P$, which is a Diffie-Hellman-type product. Without either party's secret, this reduces to ECDHP. In the ROM, directly guessing S_j so that $H_3(S_j)$ and $H_4(S_j)$ hit a specific value succeeds with probability at most $(q_{H_3} + q_{H_4})/q$.

iii) *Privacy of the payload:* Given that GK^* is hidden except with negligible probability, the challenge ciphertext pair (C^*, Tag^*) is indistinguishable by IND-CPA security of the

instantiated AEAD scheme. Hence the proposed construction achieves outsider IND-CPA confidentiality.

Moreover, in Step 3 the receiver accepts only if $M' = \text{AEAD.Dec}_{GK}(N, C, Tag; AD) \neq \perp$. Therefore, any adversary that causes a fresh ciphertext-tag pair to be accepted would violate the INT-CTXT security of the instantiated AEAD scheme. Thus chosen ciphertext queries do not provide any additional non-negligible advantage beyond the IND-CPA setting. Combining the above bounds and using the standard implication

$$\text{IND-CPA} \wedge \text{INT-CTXT} \Rightarrow \text{IND-CCA}$$

yields outsider IND-CCA confidentiality for the proposed construction [32]. $\square$

4) Authentication and the unforgeability of signatures (EUF-CMA):

[Unforgeability] An adversary $\mathcal{A}$ obtains any number of transcripts $(U, A, N, C, Tag, \delta_i)$ and makes random-oracle queries. It succeeds if it outputs a *new* $(\hat{U}, \hat{A}, \hat{N}, \hat{C}, \widehat{Tag}, \hat{\delta})$ such that with $\hat{h} = H_5(\hat{C}, \widehat{Tag}, \hat{N}, PID_i, \hat{U}, P_{pub}, \hat{A})$, the verification (8) holds for $(\hat{U}, \hat{A}, \hat{N}, \hat{C}, \widehat{Tag}, \hat{\delta})$ not previously produced by the honest sender.

Theorem 6.2 (EUF-CMA of δ_i): In the ROM, if ECDLP is hard in $\mathbb{G}$, then any PPT forger $\mathcal{A}$ has

$$\text{Adv}_{\text{DTSM}}^{\text{EUF-CMA}}(\mathcal{A}) \leq O\left(\frac{q_{H_5}}{q}\right) + \text{Adv}^{\text{ECDLP}},$$

where q_{H_5} is the number of H_5 queries. This bound ensures that an attacker cannot forge the signature δ_i , which guarantees the authenticity of the sender.

Proof sketch: The pair (U, δ_i) with $U = uP$ and $\delta_i = z_i \cdot h_i + u$ is a one-time Schnorr-like tag whose verification is the affine form (8). Here $h_i = H_5(C, Tag, N, PID_i, U, P_{pub}, A)$. Apply the Forking Lemma on H_5 to two successful forgeries that differ only in the oracle response for the same $(\hat{C}, \widehat{Tag}, \hat{N}, PID_i, \hat{U}, P_{pub}, \hat{A})$, yielding two linear equations in the unknown z_i . Solving recovers z_i (or a discrete log), contradicting ECDLP. The $O(q_{H_5}/q)$ term accounts for the small probability of oracle collisions that would invalidate the fork. This proof ensures that only an honest sender, who knows the private key associated with z_i , can generate a valid δ_i tag. Therefore, this guarantees the authenticity of the sender and prevents attackers from impersonating legitimate senders. $\square$

5) legitimacy of the receiver:

Theorem 6.3: legitimacy of the receiver]

Without knowledge of either the sender-side secret z_i or the receiver-side secret z_j (or the fresh u), a PPT adversary cannot compute any valid hidden point (α_j, γ_j) and thus cannot reconstruct f from $\{(\alpha_j, \gamma_j)\} \cup A$ to obtain $GK = f(0)$, except with negligible probability bounded by ECDHP and $(q_{H_3} + q_{H_4})/q$.

Proof sketch: By the degree bound, one missing point suffices to keep $f(0)$ undetermined; the only way to fill the gap is to know some (α_j, γ_j) . Computing S_j requires solving a DH product that is inaccessible to a passive adversary; in the ROM, guessing α_j and γ_j without S_j succeeds with probability at most $(q_{H_3} + q_{H_4})/q$. $\square$

6) Replay Resistance:

[Non-cross replay] Each execution samples a fresh u and emits a fresh $U = uP$, a fresh A , and a fresh AEAD nonce N . Moreover, $h_i = H_5(C, Tag, N, PID_i, U, P_{pub}, A)$ domain-binds the message, identity, session base U , system key, and the specific A . Thus, transcripts from other rounds or domains fail verification under (8).

Proof sketch: Replaying $(U, A, N, C, Tag, \delta_i)$ with a different (U', A', N', C', Tag') changes the input to H_5 and hence changes h_i , which breaks (8). Replaying across domains changes the H_5 input (domain separation), again failing (8). $\square$

The ephemeral scalar u is not reused between two different sessions. If the same u is reused in two sessions and their record hashes h_i are different, an attacker observing both records can compute the recovery of the long-term secret. Furthermore, since the calculation of the hidden anchor also depends on $(z_i + u)$, reusing the nonce can introduce cross session linking. Therefore, u and r must be regenerated for each new multicast instance.

7) Security Summary:

Combining the above results, and under the hardness of ECDLP and ECDHP in $\mathbb{G}$ together with the ROM assumptions for $H_1, \dots, H_6$, the DTSN confirms:

- 1) *Confidentiality:* GK is information-theoretically hidden from A alone; M follows from IND-CPA of AEAD, and further achieves IND-CCA under the AEAD properties of IND-CPA and INT-CTXT.
- 2) *Integrity/Authentication:* δ_i is EUF-CMA via a Schnorr-style proof under ECDLP.
- 3) *Legitimacy of the receiver:* Only designated MUAVs can derive (α_j, γ_j) and reconstruct GK , by the ECDHP hardness and ROM bounds.
- 4) *Replay Resistance:* Fresh (U, A, N) , hash binding preclude cross-session or cross-domain replay attacks.

C. Security Proof Based on Scyther

In this section, we employ the formal verification tool Scyther to verify the security of the scheme. Scyther is an effective formal analysis tool compatible with various security models [50]. Here, we mainly model the proposed digital twin-enabled multicast signcryption phase under the DY model.

We define three roles: HAP, MUAV and SUAV. During this phase, the HAP's public and private keys are denoted as $PK(HAP)$ and $SK(HAP)$, respectively, while the MUAV and SUAV's public and private keys are denoted as $PK(MUAV)$, $SK(MUAV)$ and $PK(SUAV)$, $SK(SUAV)$, respectively. The protocol interaction process is modeled using `send` and `recv` events. We use M, M_{agg} and GK, GK_{new} to represent the secret message and the negotiated group key, respectively. In terms of security requirements, both the HAP and MUAV must ensure the confidentiality of M and GK , the MUAV and SUAV must ensure the confidentiality of M_{agg} and GK_{new} . To verify that the protocol meets this requirement, we add two `Secret` claim events for HAP and MUAV, respectively:

Claim	Status	Comments
broadcastAuthentication HAP1	Verified	No attacks
broadcastAuthentication HAP2	Verified	No attacks
broadcastAuthentication HAP3	Verified	No attacks
broadcastAuthentication HAP4	Verified	No attacks
broadcastAuthentication HAP5	Verified	No attacks
broadcastAuthentication HAP6	Verified	No attacks
broadcastAuthentication MUAV1	Verified	No attacks
broadcastAuthentication MUAV2	Verified	No attacks
broadcastAuthentication MUAV3	Verified	No attacks
broadcastAuthentication MUAV4	Verified	No attacks
broadcastAuthentication MUAV5	Verified	No attacks
broadcastAuthentication MUAV6	Verified	No attacks
broadcastAuthentication SUAV1	Verified	No attacks
broadcastAuthentication SUAV2	Verified	No attacks
broadcastAuthentication SUAV3	Verified	No attacks
broadcastAuthentication SUAV4	Verified	No attacks
broadcastAuthentication SUAV5	Verified	No attacks
broadcastAuthentication SUAV6	Verified	No attacks

Fig. 3. Results of formal verification using the Scyther tool. Each row represents an assertion event related to HAP, MUAV, or SUAV. The “OK” and “Verified” statuses indicate that all specified properties have been successfully verified.

$\text{claim}(\text{HAP}, \text{Secret}, M), \text{claim}(\text{HAP}, \text{Secret}, GK);$
 $\text{claim}(\text{MUAV}, \text{Secret}, M), \text{claim}(\text{MUAV}, \text{Secret}, GK).$
 Furthermore, to verify mutual authentication between two roles, we introduce the `Alive` and `Weakagree` assertion events. Furthermore, we add the `Niagree` and `Nisynch` assertion events for all roles to verify the consistency of the aforementioned values and the correct synchronization of relevant messages in the protocol. And two `Secret` claim events for MUAV and SUAV. Furthermore, to verify mutual authentication between two roles, we introduce the `Alive` and `Weakagree` assertion events. Fig. 3 shows the results of verifying the digital twin-enabled multicast signcryption phase using the Scyther tool. The formal analysis results are as follows:

- 1) The fact that `Secret` is not attacked indicates that the HAP and MUAV can confirm that the secret message M and the group key GK are kept secret. The MUAV and SUAV can confirm that the secret message M_{agg} and the group key GK_{new} are kept secret.
- 2) The fact that `Alive` is not attacked indicates that all participating roles are active.
- 3) The fact that `Weakagree` is not attacked ensures that both parties in the protocol are indeed communicating. Specifically, the HAP, MUAV and MUAV, SUAV have confirmed each other's identities, thus achieving mutual authentication.
- 4) Both the `Niagree` and `Nisynch` claims are immune to attacks, ensuring the consistency of message content and order within the protocols. Similarly, our scheme uses fresh random numbers to resist replay attacks.

D. Informal Security Analysis

1) *Conditional Anonymity:* The real identity of entity U_n is hidden by the KGC during the registration process through $PID_n = ID_n \oplus H_0(Ctx_n \parallel P_{pub} \parallel sX_n)$. Since the value sX_n involves the KGC master secret s , an outside adversary cannot recover the real identity from (PID_n, Ctx_n) together with the corresponding registered public key X_n .

2) *Traceability:* If a malicious entity is detected, the KGC can reveal its real identity by computing $ID_n = PID_n \oplus$

TABLE II
COMPARISON OF COMMUNICATION OVERHEAD IN DIFFERENT SCHEMES

Scheme	m receivers	m receivers (bits)
Ours	$ G + (2m + 1) Z_q + C + Tag + N $	$512m + 1120$
[16]	$m G + 3m Z_q + m C $	$1408m$
[17]	$3m G + 2m Z_q + m C $	$2176m$
[19]	$m G + m Z_q + m C $	$896m$
[25]	$m G + m Z_q + m C $	$896m$
[26]	$2m G + m C $	$1152m$
[27]	$(2m + 2) Z_q + m C + m\log_2 C $	$647m + 512$
[28]	$(m + 1) G + (2m + 1) Z_q + m C $	$768m + 384$
[29]	$2m G + 6m Z_q $	$1792m$
[30]	$8 G + (2m + 2) Z_q + m C $	$640m + 1536$
[31]	$2 G + Z_q + m(128 + C) + T + PID $	$256m + 1456$

$H_0(Ctx_n \parallel P_{pub} \parallel sX_n)$. Therefore, the proposed scheme enables identity tracing and accountability when necessary.

Therefore, the analysis results show that our scheme can achieve the mutual authentication between the HAP and the MUAV, as well as between MUAVs and SUAVs and can resist impersonation attacks, man-in-the-middle attacks, and replay attacks, while also supporting conditional anonymity and traceability through pseudonyms.

VII. PERFORMANCE ANALYSIS

In this section, we evaluate our proposed scheme by comparing with other related schemes in terms of communication overhead and computational cost.

A. Communication Overhead

To ensure a fair comparison, we configure the protocol evaluation scenario as follows: a single sender transmits the same message to m receivers to m receivers, and the total computational cost at the sender side for transmitting these m messages is calculated. Let $|G|$ denote the size of an elliptic curve group element, $|Z_q|$ be the length of a finite field element, and $|C|$ be the length of the ciphertext message. To balance cryptographic security and computational efficiency, both plaintext and ciphertext are 128 bits long. According to the National Institute of Standards and Technology (NIST) [51], a 128-bit security strength requires an elliptic curve of approximately 256-bit order (e.g., NISTP-256), whose underlying finite field $\mathbb{F}_q$ has a 256-bit prime modulus. Therefore, the size of a finite-field element is $|Z_q| = 256$ bits. When using uncompressed elliptic-curve points, a group element consists of two field elements, and thus $|G| = 2|Z_q| = 512$ bits. The hash output length is set to 128 bits. The size of the real identity information is 128 bits, the timestamp $|T|$ is 48 bits, and the size of anonymous identifier $|PID|$ is 128 bits. In implementation, we instantiate AEAD with AES GCM. Following NIST SP 800-38D [52], we set a nonce/IV $|N| = 96$ bits and use an authentication tag $|Tag| = 128$ bits. For *Step 1* and *Step 3*, which are the HAP multicast and MUAV message processing phase: Table II compares the communication overhead of each scheme, and Fig. 4 presents a comparison of the corresponding communication overhead of messages sent by the sender in different schemes as the number of receivers increases. The results demonstrate that the proposed scheme

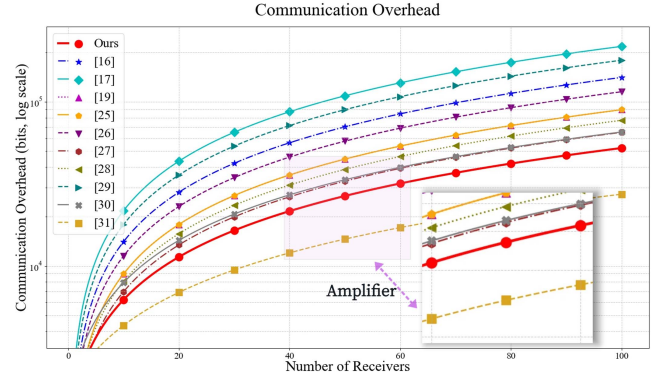


Fig. 4. Comparison of Communication Overhead in different Schemes.

exhibits significant advantages, particularly when the number of receivers is large.

B. Computational Overhead

On the computational overhead, this cryptographic performance test is conducted on the Windows 10 operating system, using an Intel 64 Family 6 Model 183 processor. The test used the Elliptic Curve Digital Signature Algorithm (ECDSA) library for elliptic curve operations, the cryptography library for symmetric cryptography operations. Each operation is executed one hundred thousand times, and the average execution time is calculated. We take the average of 10^5 calculations and specifically: elliptic curve scalar multiplication operation $T_{pm} : 0.9954$ ms, elliptic curve point addition operation $T_{pa} : 0.0259$ ms, cryptographic hash operation $T_h : 0.0028$ ms, XOR operation $T_{\oplus} : 0.0484$ ms, symmetric encryption operation $T_E : 0.0112$ ms, symmetric decryption operation $T_D : 0.0214$ ms, finite-field multiplication operation $T_* : 0.0002$ ms, finite-field addition operation $T_+ : 0.0002$ ms, finite-field inversion operation $T_{inv} : 0.1895$ ms. Then the HAP executes batched-inversion barycentric Lagrange multi-evaluation algorithm $T_{Lag}(m)$ in *Step 1*, which uses $m + 1$ interpolation points to construct a unique m -order polynomial, and then calculates and outputs the values of the polynomial at the given m query points. We adopt $T_{Lag}(m)$: given $n = m + 1$ distinct pairs $(x_i, y_i) \in \mathbb{F}_q^2$ with $x_i \neq x_j$ for $i \neq j$, we precompute barycentric weights $w_i = (\prod_{j \neq i} (x_i - x_j))^{-1}$ and $u_i = w_i y_i$, and for each query point

TABLE III
COMPUTATIONAL OVERHEAD IN DIFFERENT SCHEMES

Scheme	Sender Computational Overhead	Total (ms)
Ours	$(1 + 2m)T_{pm} + mT_{pa} + (3m + 1)T_h + T_{Lag}(m) + T_E + T_* + 2T_+$	$0.002m^2 + 2.4061m + 1.1997$
[16]	$m(3T_{pm} + T_{pa} + 3T_h + T_E + T_* + T_+)$	$3.0320m$
[17]	$m(6T_{pm} + 4T_{pa} + 5T_h + 5T_* + 3T_+ + 3T_{inv} + T_{\oplus})$	$6.7084m$
[19]	$m(6T_{pm} + 2T_{pa} + 5T_h + T_{\oplus})$	$6.0865m$
[25]	$m(5T_{pm} + 3T_{pa} + 6T_h + T_* + 3T_+ + T_{\oplus})$	$5.1204m$
[26]	$m(3T_{pm} + 2T_h + T_* + T_+ + T_{\oplus})$	$3.0405m$
[27]	$(1 + 2m)T_{pm} + mT_{pa} + (5m + 1)T_h + T_{Lag}(m) + mT_E + T_* + T_+$	$0.002m^2 + 2.4229m + 1.1883$
[28]	$(1 + 2m)T_{pm} + 2mT_{pa} + (3m + 3)T_h + m(m + 1)T_*/2 + mT_{\oplus} + T_E + (m(m + 1) + 12)T_+/2$	$0.0002m^2 + 2.0996m + 1.0162$
[29]	$m(6T_{pm} + T_{pa} + 3T_h + T_{inv} + 2T_* + T_+)$	$6.2008m$
[30]	$(3 + 2m)T_{pm} + mT_{pa} + (2m + 2)T_h + 2mT_* + mT_+$	$2.0229m + 2.9918$
[31]	$(1 + 2m)T_{pm} + 2mT_{pa} + (3m + 1)T_h + mT_{\oplus} + T_* + 3T_+$	$2.0994m + 0.9990$

TABLE IV
COMPARISON OF SUCCESS RATES AT DIFFERENT BER ($m = 10$)

Scheme	L_{avg}	10^{-5}	5×10^{-5}	10^{-4}
Ours	6240	93.95%	73.20%	53.58%
[16]	14080	86.87%	49.46%	24.46%
[17]	21760	80.44%	33.69%	11.35%
[19]	8960	91.43%	63.89%	40.82%
[25]	8960	91.43%	63.89%	40.82%
[26]	11520	89.12%	56.21%	31.60%
[27]	6982	93.26%	70.53%	49.75%
[28]	8064	92.25%	66.82%	44.64%
[29]	17920	83.59%	40.82%	16.66%
[30]	7936	92.37%	67.25%	45.22%
[31]	4352	95.74%	80.44%	64.71%

TABLE V
COMPARISON OF PROTOCOL COMPLETION DELAY UNDER DIFFERENT JITTER LEVELS ($m = 10$, $R = 1$ MBPS)

Scheme	Baseline (ms)	$J = 0$	$J = 5$	$J = 10$
Ours	31.70	31.70	36.70	41.70
[16]	44.40	44.40	49.40	54.40
[17]	88.84	88.84	93.84	98.84
[19]	69.83	69.83	74.83	79.83
[25]	60.16	60.16	65.16	70.16
[26]	41.93	41.93	46.93	51.93
[27]	32.60	32.60	37.60	42.60
[28]	30.10	30.10	35.10	40.10
[29]	79.93	79.93	84.93	89.93
[30]	31.16	31.16	36.16	41.16
[31]	25.35	25.35	31.35	36.35

$t \notin \{x_0, \dots, x_m\}$ evaluate $f(t) = (\sum_{i=0}^m \frac{u_i}{t-x_i}) / (\sum_{i=0}^m \frac{w_i}{t-x_i})$ using the standard barycentric form of Lagrange interpolation [53]; to avoid $(m + 1)$ inversions per query, we apply Montgomery's trick (batch inversion), which computes inverses of n field elements with $3(n - 1)$ field multiplications and one field inversion [54], [55]. The one-time precomputation costs $(m^2 + 5m + 1)T_* + (m^2 + m)T_+ + 1T_{inv}$, each query costs $(5m + 3)T_* + (3m + 1)T_+ + 2T_{inv}$, and thus for m queries the total cost $T_{Lag}(m) = (6m^2 + 8m + 1)T_* + (4m^2 + 2m)T_+ + (2m + 1)T_{inv}$. Therefore, the batched-inversion barycentric Lagrange multi-evaluation algorithm costs $T_{Lag}(m) : 0.002m^2 + 0.381m + 0.1897$ ms.

For *Step 1* and *Step 3*, which are the HAP multicast and MUAV message processing phase, Table III presents a comparison of the measured computational overhead at the sender side between the proposed scheme and existing schemes. Fig. 5 illustrates a comparison of the computational overhead of messages sent by the sender in different schemes as the number of receivers increases. The results indicate that the proposed scheme exhibits significant advantages, particularly in scenarios involving multiple receivers and larger receiver populations.

C. Theoretical Impact Analysis of Network Fluctuations on Protocol Performance

In real-world low-altitude wireless environments, multicast transmission can also be affected by network fluctuations,

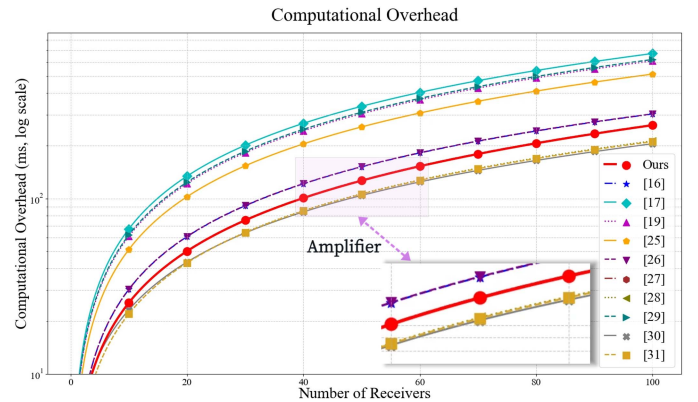


Fig. 5. Comparison of Computational Overhead in different Schemes.

especially link unreliability and latency jitter. We will further discuss the impact of these two factors on protocol performance.

1) *Effect of Link Unreliability on Message Success Rate:* HAP sends a one-time multicast message to the designated MUAV, and each receiver then reconstructs the group key, decrypts the ciphertext, and verifies the sender's tag. Therefore, the most impact of wireless link instability is the success rate of message reception.

Let ε denote the bit error rate (BER) of the wireless link, and L_{avg} denote the total number of transmitted bits in the

TABLE VI
COMPARISON OF VARIOUS SCHEMES

Properties	[16]	[17]	[19]	[25]	[26]	[27]	[28]	[29]	[30]	[31]	Ours
Confidentiality	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Integrity	✓	✓	✓	✓	✓	✓	×	×	✓	✓	✓
Authentication	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Unforgeability	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Replay Resistance	×	✓	✓	✓	✓	✓	×	✓	✓	✓	✓
MITM/Impersonation Resistance	✓	✓	✓	✓	✓	✓	×	✓	✓	✓	✓
Conditional Anonymity	✓	×	×	×	×	✓	✓	✓	✓	✓	✓
Traceability	✓	×	✓	✓	✓	✓	×	✓	×	✓	✓
Single-Signcryption for Multiple Receivers	×	×	×	×	×	×	×	×	×	✓	✓
AEAD-based Decryption Validity Check	×	×	×	×	×	×	×	×	×	×	✓
AD-Based Ciphertext Context Binding	×	×	×	×	×	×	×	×	×	×	✓
INT-CTXT	×	×	×	×	×	×	×	×	×	×	✓
Trusted Relay Forwarding	×	×	×	×	×	×	×	×	×	×	✓
Dynamic Group Update	×	×	×	×	×	×	×	×	×	×	✓
Forwarding Multi-task Aggregation	×	×	×	×	×	×	×	×	×	×	✓
Communication Overhead	High	High	Middle	Middle	High	Middle	Middle	High	Middle	Low	Low
Computational Overhead	Middle	High	High	High	Middle	Low	Low	High	Low	Low	Low

direct-delivery phase. Under the independent bit-error model, the direct-delivery success rate can be approximated by $P_{\text{succ}}^{\text{dir}} = (1 - \varepsilon)^{L_{\text{avg}}}$. Hence, under the same BER, a scheme with a shorter transcript is less sensitive to link unreliability. To provide an intuitive comparison, we fix the receiver size as $m = 10$ and use the communication-overhead expressions as shown in Table II to evaluate the resulting success rate under three representative BER levels, namely, $\varepsilon = 10^{-5}$, 5×10^{-5} , and 10^{-4} , which correspond to low, medium, and relatively severe link-error conditions, respectively. The results are shown in Table IV.

As shown in Table IV, at the same bit error rate (BER), the transmission success rate depends primarily on the total number of bits transmitted. Because the proposed scheme has a relatively short transmission time, its message success rate is higher than most existing schemes on unreliable links. The scheme in [31] achieves an even shorter direct transcript under the unified evaluation setting, and therefore attains a higher direct-delivery success rate than our scheme.

2) *Impact of Jitter on Protocol Completion Delay*: The multicast transmission is also affected by jitter. Jitter does not change the inherent encryption complexity of the protocol, but it increases the uncertainty on the network side during the message transmission. Protocol completion delay can be expressed as: $T_{\text{dir}} = T_{\text{comp}} + \frac{L_{\text{avg}}}{R} + J$, where T_{comp} represents the protocol computation overhead delay obtained from the Table III, L_{avg} represents the protocol communication overhead, R is the effective link transmission rate, and J represents the additional delay introduced by jitter.

For numerical illustration, we fix $m = 10$ and set $R = 1$ Mbps as the nominal rate to convert the communication overhead into transmission delay. In addition, we consider the following three representative jitter levels: $J = 0, 5$ ms, and 10 ms, corresponding to no jitter, moderate jitter, and significant jitter, respectively. The final protocol completion delay is shown in Table V.

As shown in Table V, jitter has an approximately linear increase on the completion delay of all schemes, and the relative

ranking of the protocols mainly depends on the baseline communication and computational overhead of the scheme itself. Due to the low baseline delay, the proposed scheme can still maintain good delivery delay under jitter, and its performance is close to the schemes in [28] and [30], and better than most other comparative schemes. This shows that the proposed scheme can still maintain good link unreliability robustness and good delay performance under jitter. We note that the message success rate and protocol completion delay of the scheme in [31] are both slightly better than those of our proposed scheme. However, based on the multi-dimensional security function and attribute evaluation in Section VII-D, our security attributes are much stronger and our security functions are more numerous than the scheme in [31]. Overall, our proposed scheme delivers stronger security capabilities while achieving comparable performance levels.

D. Discussion

Our scheme aims to substantially reduce the communication and computational overhead on the sender side by optimizing the process through a polynomial interpolation mechanism. Compared with most comparative schemes, the proposed design achieves lower sender-side overhead in the considered multi-receiver setting. As can be seen from Table VI, our scheme not only provides robust security properties including confidentiality, integrity, authentication, unforgeability, several protocol attacks resistance, conditional anonymity, traceability and single-signcryption for multiple receivers, but our scheme can also achieve the functions of AEAD-based decryption validity check, AD-based ciphertext context binding, INT-CTXT, trusted relay forwarding, dynamic group update and forwarding multi-task aggregation compared to all the other schemes. Only one signcryption is generated for the same task, and all designated receivers share the same group key to decrypt and verify the signature, which greatly reduces communication overhead and

is suitable for large-scale UAVs to perform the same task. Our scheme has similar computational overhead to these schemes in [28], [30] and [31], which is much better than other schemes in [16], [17], [19], [25], [26], [27], [29]. But compared to these schemes, our scheme achieves more security attributes and functions as shown in Table VI. And compared to the scheme in [31], our confidentiality design is based on an AEAD construction, which provides a more direct path to an IND-CCA claim. By contrast, the CCA2 proof in [31] is presented under a simplified abstraction of the signcryption / unsigncryption process, the correspondence between its proof model and the full protocol construction may benefit from further clarification. Therefore, considering both security features and performance, our scheme is more suitable for large-scale UAVs collaborative scenarios.

VIII. CONCLUSION

To address the “receiver-friendly but sender-heavy” structural imbalance in UAV swarm cooperative communication, this paper proposes a multicast signcryption protocol driven by DTs. We construct a three-layer digital twin architecture to achieve dynamic resource scheduling and precise virtual-physical mapping. Furthermore, a key distribution mechanism based on Lagrange interpolation is designed, where the sender embeds the group key into the constant term of a constructed polynomial. This approach substantially reduces sender-side computational and communication overhead in large-scale multicast scenarios.

Formal security analysis under the random oracle model confirms the confidentiality and unforgeability of the scheme, while formal verification using Scyther validates its resistance to replay and man-in-the-middle attacks. Ultimately, this work demonstrates that leveraging DTs for receiver orchestration offers an efficient solution for 6G low-altitude networks. In this paper, we employ the Dolev-Yao model as the threat model, which primarily focuses on protocol/message layer manipulation. Therefore, physical layer attacks such as radio jamming, and implementation-level information leakage (e.g., side-channel attacks) are not considered in this paper. Future research will explore the integration of quantum-safe primitives, extending this approach to cross-domain heterogeneous networks, and investigating its resilience under stronger threat models that incorporate both physical layer attacks and implementation-level security.

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